

LAB-14: Analysis of Activation Functions and Optimizers for Neural Network Model

Course- B. Tech.	Type- Core
Course Code- CSET301	Course Name- Artificial Intelligence and Machine Learning
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CO-Mapping

	C01	C02	C03	C04	C05	C06
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Objectives

- Understand the purpose of activation functions in neural networks, particularly how they introduce non-linearity and impact learning.
- Compare commonly used activation functions such as ReLU, tanh, and logistic in terms of accuracy and training behavior.
- Learn how optimizers like Adam and SGD affect convergence speed, stability, and overall model performance.
- Train the model with different activation-optimizer combinations and observe differences in accuracy and convergence.
- Understand which combinations are better suited for faster learning and which may perform poorly.
- Develop the ability to critically analyze why certain activation-optimizer pairs outperform others on a given dataset.

- Document your results in a structured format and summarize which activation and optimizer pair works best and why.

Steps to Solve

- Load the Breast Cancer dataset from scikit-learn.
- Split the dataset into training and testing sets.
- Standardize the dataset using StandardScaler.
- Select different activation functions (ReLU, tanh, logistic) and optimizers (Adam, SGD) for experimentation.
- Train the neural network with MLPClassifier for each combination of activation function and optimizer.
- Record the accuracy for each run and compare the results.
- Write a conclusion explaining which activation-optimizer pair is most effective and why.